One-Dimensional Slug-Flow Model for Self-Starting Siphons: Theory, Bounds, and Computation

IYPT 2026 Problem #1

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Abstract

Self-starting siphons prime spontaneously when partially immersed, without external suction. This document develops a one-dimensional slug-flow model that connects geometric and physical parameters to observable outcomes—priming success, filling time, and draining rate—for IYPT 2026 Problem #1.

The model yields three main results: (i) an analytical upper bound $h_{\max}$ for the maximum climb height, expressed in closed form for piecewise-linear pipe geometries; (ii) a geometric crest-crossing criterion $h_{\max} > h_{\text{crest}}$ that rapidly screens infeasible designs without numerical integration; and (iii) explicit ordinary differential equations for the full time evolution, incorporating Churchill friction and minor losses.

The deliberate simplicity of treating the liquid as a rigid slug enables rapid parameter exploration (seconds per case) and makes the physics transparent. Discrepancies between model predictions and experiments or CFD directly highlight where additional mechanisms—such as three-dimensional flow structure, air entrainment, or column separation—become important.

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1 Introduction

1.1 The IYPT problem

The International Young Physicists' Tournament (IYPT) 2026 Problem #1 states:

> *"A self-starting siphon can be made using a piece of rigid tubing bent into a specific shape. When the siphon is partially immersed in water, it begins siphoning water without the need for initial suction. Investigate how the relevant parameters, such as the geometry, affect the siphoning process."*

This open-ended question requires identifying the controlling parameters, establishing quantitative relationships between parameters and outcomes, understanding underlying mechanisms and their limits, and exploring failure regimes.

1.2 Physical mechanism

Self-starting siphons differ from conventional inverted-U siphons in that they do not require prefilling or an externally imposed pressure difference. Instead, the geometry is arranged so that an initial drop converts gravitational potential energy into kinetic energy, and the resulting momentum carries a water slug over a crest before friction and gravity bring it to rest. Once the crest is crossed, standard siphon action continues until the driving head is exhausted. Classical and teaching-oriented discussions of siphon behaviour and the role of atmospheric pressure and cohesion can be found in Potter and Barnes [6], Jumper and Stanchev [8], and Hughes [7].

The key insight is that in a properly designed self-starting geometry, the driving head $h_c(t) - h(L(t))$ (container surface minus front elevation) remains positive for an extended portion of the climb, allowing the slug to build momentum. By contrast, in a simple inverted-U siphon the driving head vanishes at the crest after the initial drop, so no momentum advantage accumulates.

1.3 Document structure

Section 2 establishes the geometry and notation. Section 3 derives volume conservation and the container drawdown relation. Section 4 presents the unified momentum derivation from first principles. Section 5 assembles the complete filling-phase dynamics. Section 6 develops the analytical climb bound and geometric priming criterion. Section 7 treats the draining phase. Section 8 discusses validity limits. Section 9 presents the computational implementation, and Section 11 summarises the model.

2 Problem Setup: Geometry and Notation

2.1 Coordinate system

We work in a two-dimensional vertical plane with coordinates (x, h), where x is horizontal (increasing rightward) and h is vertical (increasing upward). The vertical datum is chosen so that $h = 0$ at the container base.

2.2 Pipe geometry: five nodes and four segments

The pipe centreline is defined by five nodes

$$A = (x_0, h_0), \quad B = (x_1, h_1), \quad C = (x_2, h_2), \quad D = (x_3, h_3), \quad E = (x_4, h_4),$$

connected by four straight segments. Node A is at the pipe inlet (container wall), and node E is at the outlet. By convention, D is the crest (highest point on the path).

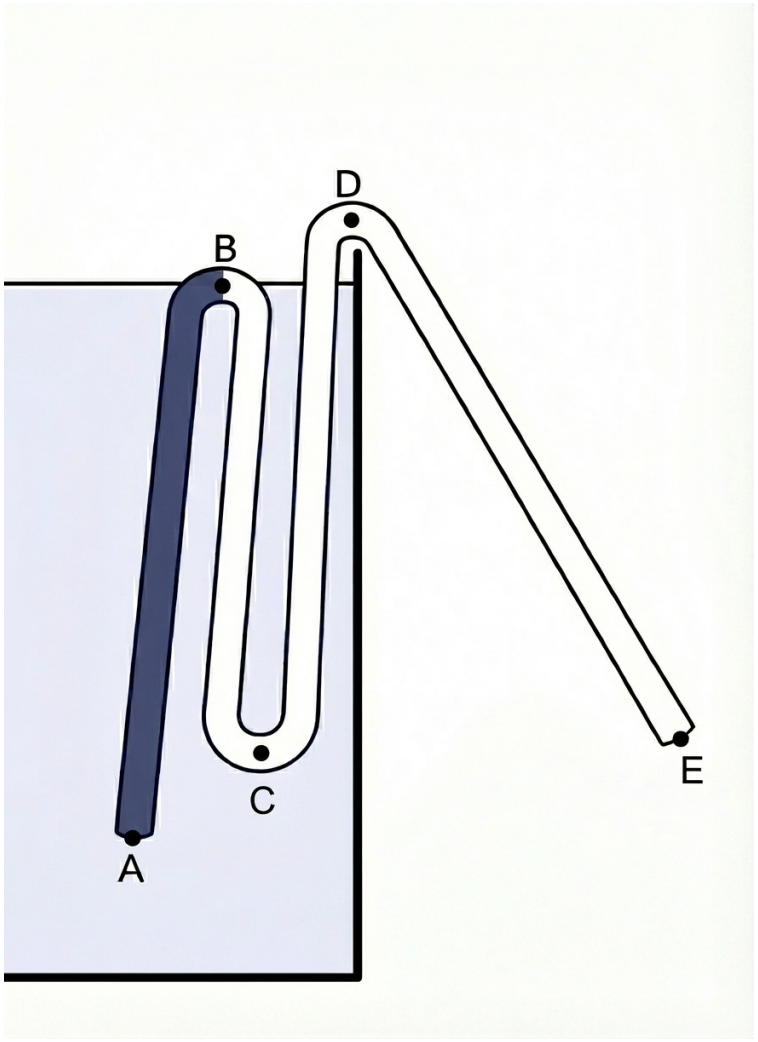


Figure 1: Schematic of the self-starting siphon geometry and node labelling.

The length of each segment is

$$L_i = \sqrt{(x_i - x_{i-1})^2 + (h_i - h_{i-1})^2}, \qquad i = 1, 2, 3, 4,$$

and the total pipe length is $L_{\max} = L_1 + L_2 + L_3 + L_4$.

2.3 Arclength coordinate and centreline elevation

We parameterise the pipe centreline by arclength L, measured from the inlet:

$$L = 0 \text{ at } A, \quad L = L_1 \text{ at } B, \quad L = L_1 + L_2 \text{ at } C, \quad L = L_1 + L_2 + L_3 \text{ at } D, \quad L = L_{\max} \text{ at } E.$$

It is convenient to define cumulative lengths

$$S_0 = 0, \quad S_1 = L_1, \quad S_2 = L_1 + L_2, \quad S_3 = L_1 + L_2 + L_3, \quad S_4 = L_{\max}.$$

Since each segment is straight, the centreline elevation $h(L)$ is piecewise linear. On segment i (for $L \in [S_{i-1}, S_i]$) we write

$$h(L) = c_i + s_i L, \tag{1}$$

where the slope is

$$s_i = \frac{h_i - h_{i-1}}{L_i} = \sin\theta_i \tag{2}$$

(with θ_i the inclination angle to horizontal) and the intercept is

$$c_i = h_{i-1} - s_i S_{i-1}. \tag{3}$$

Only the vertical projection of each leg enters the dynamics—horizontal extent affects segment length but not the gravitational driving force.

2.4 State variables

During the *filling phase*, the pipe is partially full. The state is characterised by:

- $L(t)$: the wetted length (arclength from inlet to the advancing front),
- $h_c(t)$: the container free-surface elevation.

The front velocity is $v(t) = \dot{L}(t)$, and the front elevation is $h(L(t))$.

During the *draining phase*, the pipe is completely full ($L = L_{\max}$), and the state becomes $(h_c(t), v(t))$ where $v(t)$ is now the mean pipe velocity.

2.5 Container and pipe parameters

Symbol	Units	Description
A_0	m^2	Effective container cross-sectional area
A	m^2	Pipe cross-sectional area, $A = \pi D^2/4$
D	m	Pipe inner diameter
ρ	$\mathrm{kg/m}^3$	Fluid density
μ	Pa·s	Dynamic viscosity
ε/D	—	Relative pipe roughness
$\sum K_i$	—	Total minor-loss coefficient

The container is assumed large enough that the free surface remains quasi-hydrostatic (velocities at the surface are negligible compared to pipe velocities).

3 Mass Conservation and Container Drawdown

3.1 Volume balance during filling

During the filling phase, no water has left the outlet, so the total volume in the container–pipe system is conserved:

$$V_{\mathrm{cont}}(t) + V_{\mathrm{pipe}}(t) = V_{\mathrm{tot}} = \text{constant}. \tag{4}$$

With $V_{\mathrm{cont}}(t) = A_0 h_c(t)$ and $V_{\mathrm{pipe}}(t) = AL(t)$, this becomes

$$A_0 h_c(t) + AL(t) = A_0 h_c(0) + AL(0).$$

3.2 Initial conditions

We define $t = 0$ as the instant when the dynamical filling phase begins. Two common preparation procedures lead to the same initial state:

1. *Pre-fill*: slowly fill the container until the water surface reaches node B, fully wetting segment AB.
2. *Dip*: submerge the inlet until the internal level reaches B, then position the device.

In both cases, the preparatory motion is slow, so at $t = 0$ the front has just arrived at B with negligible velocity. The initial conditions are therefore

$$\boxed{L(0) = L_1, \qquad \dot{L}(0) = 0, \qquad h_c(0) = h_0.} \tag{5}$$

The total volume is $V_{\text{tot}} = A_0 h_0 + A L_1$.

3.3 Container level as a function of wetted length

Solving (4) for h_c in terms of L gives the key kinematic relation:

$$\boxed{h_c(L) = h_0 - \frac{A}{A_0}(L - L_1).} \tag{6}$$

The container level decreases linearly as the front advances. The ratio A/A_0 controls how quickly the driving head diminishes: large containers ($A_0 \gg A$) experience minimal drawdown, while small containers can lose driving head rapidly.

Remark on generalisation. For complex internal pipe layouts where the number of pipe legs intersected by a horizontal slice varies with elevation, the effective area $A_{\text{eff}}(\eta)$ becomes height-dependent. The volume map is then $V_{\text{cont}} = \int_0^{h_c} A_{\text{eff}}(\eta)\, d\eta$, and $h_c(L)$ must be obtained from an implicit relation. In this document, we adopt the constant-A_0 approximation, which is accurate when the pipe diameter is small compared to the container radius.

4 Momentum Balance: Derivation from First Principles

This section derives the equation of motion for the water slug by integrating pressure and gravitational forces along the pipe. The derivation is presented once, in full, to establish the result rigorously.

4.1 Setup

Consider the water in the pipe as a slug of length $L(t)$, cross-sectional area A, and uniform mean velocity $v(t) = \dot{L}(t)$. Introduce a local arclength coordinate ξ measured from the inlet, so ξ runs from 0 (at A) to $L(t)$ (at the front).

At position ξ, the centreline elevation is $h(\xi)$ and the pressure is $p(\xi, t)$.

4.2 Forces on a fluid slice

Consider a thin slice between ξ and $\xi + d\xi$:

- Mass: $dm = \rho A\, d\xi$
- Upstream pressure force (in direction of motion): $+p(\xi, t)A$
- Downstream pressure force: $-p(\xi + d\xi, t)A$

- Gravitational component along pipe: $-\rho g A \dfrac{dh}{d\xi}\, d\xi$

The net tangential force on the slice is

$$dF = -A\frac{\partial p}{\partial \xi}\, d\xi - \rho g A \frac{dh}{d\xi}\, d\xi.$$

4.3 Integration along the slug

Integrating from $\xi = 0$ to $\xi = L$ gives the total force:

$$\begin{aligned} F_{\text{net}} &= -A\int_0^L \frac{\partial p}{\partial \xi}\, d\xi - \rho g A \int_0^L \frac{dh}{d\xi}\, d\xi \\ &= A\big[p(0,t) - p(L,t)\big] + \rho g A\big[h(0) - h(L)\big]. \end{aligned} \tag{7}$$

4.4 Boundary pressures

We now apply the boundary conditions:

- **Inlet**: The container is hydrostatic with free surface at $h_c(t)$ and atmospheric pressure p_{atm}. At the inlet (elevation $h(0) = h_0$),

$$p(0,t) = p_{\text{atm}} + \rho g\big[h_c(t) - h_0\big].$$

- **Front**: During filling, the front is a free surface at atmospheric pressure,

$$p(L,t) = p_{\text{atm}}.$$

Substituting into (7):

$$\begin{aligned} F_{\text{net}} &= A\Big[\big(p_{\text{atm}} + \rho g(h_c - h_0)\big) - p_{\text{atm}}\Big] + \rho g A\big[h_0 - h(L)\big] \\ &= \rho g A(h_c - h_0) + \rho g A(h_0 - h(L)) \\ &= \rho g A\big[h_c(t) - h(L(t))\big]. \end{aligned} \tag{8}$$

$$\boxed{F_{\text{drive}} = \rho g A\big[h_c(t) - h(L(t))\big]} \tag{9}$$

This is the fundamental result: the net driving force equals the hydrostatic pressure difference corresponding to the head between the container surface and the front. The inlet elevation h_0 cancels algebraically, showing that only the relative positions of the free surfaces matter.

4.5 Lossless equation of motion

Applying Newton's second law to the slug of mass $m(t) = \rho A L(t)$:

$$\rho A L(t)\ddot{L}(t) = \rho g A\big[h_c(t) - h(L(t))\big].$$

Cancelling ρA and substituting the drawdown relation (6):

$$\boxed{\ddot{L} = \frac{g}{L}\left[h_0 - \frac{A}{A_0}(L - L_1) - h(L)\right]} \tag{10}$$

This is the lossless equation of motion for the filling phase. The factor $1/L$ arises because the same driving force must accelerate an ever-lengthening slug.

5 Filling Phase: Complete Dynamics with Losses

5.1 Loss models

Real pipes dissipate energy through wall friction and local disturbances.

Distributed friction (Darcy–Weisbach). The head loss along a pipe of length ℓ with mean velocity v is

$$h_f = f\frac{\ell}{D}\frac{v^2}{2g},$$

where f is the Darcy friction factor [1]. For laminar flow ($\mathrm{Re} < 2000$), $f = 64/\mathrm{Re}$. For all regimes, the Churchill correlation [2] provides an explicit expression:

$$f = 8\left[\left(\frac{8}{\mathrm{Re}}\right)^{12} + \frac{1}{(A_1 + A_2)^{3/2}}\right]^{1/12}, \tag{11}$$

$$A_1 = \left[-2.457\ln\left(\left(\frac{7}{\mathrm{Re}}\right)^{0.9} + 0.27\frac{\varepsilon}{D}\right)\right]^{16}, \tag{12}$$

$$A_2 = \left(\frac{37530}{\mathrm{Re}}\right)^{16}, \tag{13}$$

where $\mathrm{Re} = \rho|v|D/\mu$ is the Reynolds number and ε/D is the relative roughness. This correlation smoothly spans laminar, transitional, and turbulent regimes and agrees with the implicit Colebrook–White equation [3] to within a few percent; see also Moody [4] for graphical validation.

Minor losses. Bends, entrances, and other fittings introduce head losses of the form

$$h_m = K\frac{v^2}{2g},$$

where K is a dimensionless loss coefficient. Standard values are tabulated in engineering references such as Crane [5].

Total head loss. For a slug of length L moving at velocity v, the total head loss is

$$h_{\mathrm{loss}}(v, L) = \left(f\frac{L}{D} + \sum_i K_i\right)\frac{v^2}{2g}. \tag{14}$$

The corresponding resistive force (opposing motion) is

$$F_{\mathrm{loss}} = \rho g A\, h_{\mathrm{loss}} \cdot \mathrm{sgn}(v) = \frac{\rho A}{2}\left(f\frac{L}{D} + \sum_i K_i\right)|v|v.$$

(We use $|v|v$ to encode direction, which is equivalent to $v^2\mathrm{sgn}(v)$.)

5.2 Extended equation of motion

Including losses, Newton's second law becomes

$$\rho A L\ddot{L} = \rho g A\big[h_c - h(L)\big] - \frac{\rho A}{2}\left(f\frac{L}{D} + \sum_i K_i\right)|\dot{L}|\dot{L}.$$

Cancelling ρA and substituting (6):

$$\boxed{\ddot{L} = \frac{g}{L}\left[h_0 - \frac{A}{A_0}(L - L_1) - h(L)\right] - \frac{1}{2L}\left(f\frac{L}{D} + \sum_i K_i\right)|\dot{L}|\dot{L}} \tag{15}$$

This is the complete filling-phase ODE. The initial conditions are (5), and the filling phase ends when $L(t^*) = L_{\max}$.

5.3 Characteristic scales

It is instructive to identify the natural scales of the problem:

- Length scale: $L_{\max}$ or total drop height Δh
- Velocity scale: $v_0 \sim \sqrt{g\Delta h}$ (free-fall estimate)
- Time scale: $\tau \sim \sqrt{L_{\max}/g}$

Dimensionless groups controlling the dynamics include A/A_0 (drawdown rate), $fL_{\max}/D + \sum K_i$ (total loss coefficient), and the Reynolds number $\mathrm{Re} = \rho v_0 D/\mu$.

6 Analytical Climb Bound and Priming Criterion

The full ODE (15) requires numerical integration. However, a deliberately optimistic idealisation yields an analytical upper bound on how high the front can climb, providing a rapid screening tool for candidate geometries.

6.1 The ideal model

We make two simplifying assumptions:

1. **No drawdown**: $h_c(t) \equiv h_0$ (infinitely large container).
2. **No losses**: $f = 0$ and $\sum K_i = 0$.

Under these assumptions, the equation of motion reduces to

$$\ddot{L} = \frac{g}{L}\big[h_0 - h(L)\big]. \tag{16}$$

The initial conditions remain $L(0) = L_1$ and $\dot{L}(0) = 0$.

6.2 Energy integral

Define the velocity as a function of position: $v(L) = \dot{L}$. Using the chain rule,

$$\ddot{L} = \frac{dv}{dt} = \frac{dv}{dL}\frac{dL}{dt} = v\frac{dv}{dL}.$$

Substituting into (16):

$$v\frac{dv}{dL} = \frac{g}{L}\big[h_0 - h(L)\big].$$

Integrating from L_1 (where $v = 0$) to a general position L:

$$\frac{v(L)^2}{2} = g \int_{L_1}^{L} \frac{h_0 - h(\xi)}{\xi}\, d\xi. \tag{17}$$

Physical interpretation. The left side is the kinetic energy per unit mass (expressed as velocity head). The integral on the right accumulates the "excess head" $h_0 - h(\xi)$ weighted by $1/\xi$. Early sections of the path (small ξ, low elevation) contribute strongly; sections near or above h_0 contribute little or are negative.

6.3 Stall condition and maximum climb height

The front stalls (comes to rest) at the first position L_{stall} where the accumulated integral returns to zero:

$$\boxed{\int_{L_1}^{L_{\text{stall}}} \frac{h_0 - h(\xi)}{\xi}\, d\xi = 0.} \tag{18}$$

In other words, L_{stall} is the first location along the pipe (if it exists) at which the front velocity v has decreased back to zero.

It is convenient to distinguish between:

- the *stall position* L_{stall}, where $v = 0$;
- the *trajectory peak* L_{peak}, where the front reaches its highest elevation along the ideal trajectory;
- the corresponding *maximum elevation*

$$h_{\max} = h(L_{\text{peak}}). \tag{19}$$

For the piecewise-linear geometry used here, node D is the global crest, so the centreline elevation $h(L)$ attains its geometric maximum at arclength $L_3 = S_3$. The highest elevation actually reached by the front during the ideal filling phase is therefore

$$L_{\text{peak}} = \min\{L_{\text{stall}},\, L_3\}, \qquad h_{\max} = h(L_{\text{peak}}). \tag{20}$$

If the stall occurs before the crest ($L_{\text{stall}} < L_3$), then $L_{\text{peak}} = L_{\text{stall}}$ and the front never reaches D. If the stall occurs at or after the crest ($L_{\text{stall}} \geq L_3$), the trajectory peak coincides with the crest ($L_{\text{peak}} = L_3$) and the front does pass D before decelerating on the downstream leg. In both cases, because the ideal model ignores drawdown and losses, $h_{\max}$ is an *upper bound* on the climb achievable in reality.

6.4 Closed-form evaluation for piecewise-linear $h(L)$

Using the affine representation (1), the contribution of a subinterval $[a, b] \subseteq [S_{i-1}, S_i]$ on segment i is

$$\begin{aligned}\int_a^b \frac{h_0 - h(\xi)}{\xi}\, d\xi &= \int_a^b \frac{h_0 - c_i - s_i\xi}{\xi}\, d\xi \\ &= (h_0 - c_i) \ln \frac{b}{a} - s_i(b - a).\end{aligned} \tag{21}$$

Define the segment contribution function:

$$\Phi_i(a, b) = (h_0 - c_i) \ln \frac{b}{a} - s_i(b - a). \tag{22}$$

For a candidate position L on segment k (i.e., $L \in [S_{k-1}, S_k]$), the total integral is

$$I(L) = \sum_{i=2}^{k-1} \Phi_i(S_{i-1}, S_i) + \Phi_k(S_{k-1}, L). \tag{23}$$

(Note: the sum starts at $i = 2$ because $L_1 = S_1$ and the integral starts at L_1.)

The stall position L_{stall} is found by locating the segment k where $I(L)$ first becomes negative, then solving

$$\sum_{i=2}^{k-1} \Phi_i(S_{i-1}, S_i) + \Phi_k(S_{k-1}, L) = 0 \tag{24}$$

for L on that segment. Since $\Phi_k(S_{k-1}, L)$ is monotonic in L, this is a simple root-finding problem (bisection or Newton's method suffices).

6.5 Geometric priming criterion

Let h_3 denote the crest elevation (at node D) and recall that, in the ideal model, the container free surface is fixed at $h_c \equiv h_0$. Define the energy integral

$$I(L) = \int_{L_1}^{L} \frac{h_0 - h(\xi)}{\xi}\, d\xi, \tag{25}$$

so that (17) reads $v(L)^2/2 = g\, I(L)$.

In the ideal model, the front reaches some position $L > L_1$ without stalling if and only if $I(L) > 0$ on $(L_1, L]$. A *necessary and sufficient* ideal priming condition (front reaches the outlet without ever coming to rest) is therefore

$$I(L) > 0 \quad \text{for all } L \in (L_1, L_{\max}]. \tag{26}$$

Equivalently, the minimum of $I(L)$ over $(L_1, L_{\max}]$ must be strictly positive.

For typical self-starting geometries, the outlet E lies below the container level ($h_4 < h_0$), and the pipe has a single global crest at D. In that case, the path crosses the container level twice: once on the up-leg between B and D, and once on the down-leg between D and E. Let L_{cross} denote the *last* arclength where $h(L_{\text{cross}}) = h_0$ (the downward crossing). For $L > L_{\text{cross}}$ one has $h(L) < h_0$ and hence $\ddot{L} > 0$: the front is strictly accelerated toward the outlet. The first stall, if it occurs at all, must therefore lie in $[L_1, L_{\text{cross}}]$, and the ideal priming condition (26) can be restricted to that interval.

While (26) is exact, it is analytically more involved to enforce than a simple geometric condition. A convenient *necessary* screening criterion is obtained by comparing the ideal maximum climb height $h_{\max}$ from (20) with the crest elevation. A necessary condition for self-priming is

$$\boxed{h_{\max} > h_3.} \tag{27}$$

If $h_{\max} \le h_3$, then even the most optimistic model predicts that the front stalls below the crest—the geometry cannot self-prime regardless of friction or wetting details. Conversely, geometries with $h_{\max} > h_3$ merely pass this preliminary check: they still require verification of the stronger condition (26) (or full ODE integration) to guarantee priming.

6.6 Utility of the bound

The ideal bound serves three purposes:

1. **Rapid screening**: Any design with $h_{\max} \leq h_3$ can be rejected without solving the full ODE.
2. **Parameter insight**: Since $h_{\max}$ depends only on geometric data $\{h_i, L_i\}$ and the ideal stall location L_{stall}, one can analytically study how geometry affects priming potential.
3. **Diagnostic**: The gap between $h_{\max}$ and the actual climb (from simulation or experiment) quantifies how strongly non-ideal effects influence performance.

7 Draining Phase Dynamics

Once the front reaches the outlet, the pipe becomes completely full and the system transitions to draining. The wetted length is now fixed at $L(t) = L_{\max}$ for $t \geq t^*$.

7.1 State variables

In the draining phase, the state is $(h_c(t), v(t))$, where:

- $h_c(t)$: container free-surface elevation,
- $v(t)$: mean pipe velocity.

The outlet elevation is $h_{\text{out}} = h(L_{\max}) = h_4$.

7.2 Mass balance

The volumetric flow rate leaving the container equals the pipe discharge:

$$\frac{dV_{\text{cont}}}{dt} = -Av(t).$$

With $V_{\text{cont}} = A_0 h_c$:

$$\boxed{\dot{h}_c = -\frac{A}{A_0} v.} \tag{28}$$

Note that total volume is no longer conserved: water is now leaving the system through the outlet. Hence h_c becomes an independent state variable.

7.3 Momentum balance

For a completely full pipe, the momentum equation is

$$\rho A L_{\max} \dot{v} = \rho g A \big[h_c(t) - h_{\text{out}}\big] - \rho g A \, h_{\text{loss}}(v, L_{\max}) \cdot \operatorname{sgn}(v).$$

Cancelling ρA:

$$\boxed{\dot{v} = \frac{g}{L_{\max}} \big[h_c - h_{\text{out}}\big] - \frac{1}{2L_{\max}} \left(f \frac{L_{\max}}{D} + \sum_i K_i \right) |v| v} \tag{29}$$

7.4 Initial conditions for draining

The draining phase begins at $t = t^*$ with:

$$v(t^*) = \dot{L}(t^*) \quad \text{(velocity at end of filling)}, \tag{30}$$

$$h_c(t^*) = h_0 - \frac{A}{A_0}(L_{\max} - L_1). \tag{31}$$

Equations (28) and (29) form a coupled ODE system that is integrated until either h_c drops to h_{out} (driving head exhausted) or v falls to zero.

8 Validity Limits

The slug model rests on several assumptions. This section quantifies when each may fail.

8.1 Modelling assumptions

Assumption 8.1 (Incompressible Newtonian fluid)**.** Constant density ρ and viscosity μ (e.g., water at room temperature).

This holds extremely well for clean water in typical IYPT siphons: density variations with temperature are at the percent level, and non-Newtonian effects only arise with very viscous or particulate fluids (which are outside the scope of this model).

Assumption 8.2 (Rigid uniform circular pipe)**.** Diameter D is constant, and the pipe does not deform under the internal pressures encountered.

Commercial plastic or glass tubing with millimetre-scale wall thickness behaves essentially rigidly over the pressure range of a few metres of water head; very soft silicone tubes can expand noticeably, effectively changing A and the friction factor.

Assumption 8.3 (Quasi-hydrostatic container)**.** The free-surface velocity is negligible: $|\dot{h}_c| \ll \sqrt{gD}$ and $|\ddot{h}_c| \ll g$.

In practice, this requires a container cross-section A_0 much larger than the pipe area A so that $h_c(t)$ changes slowly. Narrow supply tubes or very small reservoirs violate this assumption and would need a more detailed container-flow model.

Assumption 8.4 (Single connected slug)**.** The water column remains continuous with nearly uniform velocity across each cross-section.

Once the column breaks into multiple slugs separated by air (for example after strong cavitation or vigorous inlet entrainment), the present one-slug model no longer applies and a multi-slug or two-phase description would be required.

8.2 Cavitation

Along the centreline, especially near the crest, the absolute pressure must remain above the vapour pressure p_{vap} to avoid column separation. In terms of head:

$$h_c - h_{\text{crest}} - h_{\text{loss,crest}} \geq h_{\text{vap}}, \qquad h_{\text{vap}} = \frac{p_{\text{atm}} - p_{\text{vap}}}{\rho g} \approx 10.3\,\text{m}. \tag{32}$$

Violation of (32) indicates that the modelled negative gauge pressures cannot be sustained; vapour bubbles or air would appear, breaking the slug. Modern discussions of siphon height limits and negative pressures include Boatwright et al. [16] and Vera et al. [17]; for broader background on cavitation and water under tension, see Caupin and Herbert [18], and the role of water under tension in plant xylem is reviewed by Tyree [19]. For tabletop siphons with crest heights of order 1 m the left-hand side of (32) is typically many metres larger than h_{vap}, so cavitation is unlikely; the inequality becomes important only for very tall devices or unusually large losses between the container and the crest.

8.3 Surface tension and capillarity

For small diameters, the meniscus at the inlet can sustain a finite pressure jump. The Jurin height is approximately [1]

$$h_\sigma \approx \frac{4\sigma\cos\theta}{\rho g D},$$

where σ is the surface tension and θ is the contact angle. Capillary effects are negligible when the Bond number

$$\mathrm{Bo} = \frac{\rho g D^2}{\sigma} \gg 1.$$

For water at room temperature with $D = 5\,\mathrm{mm}$, $\mathrm{Bo} \approx 3.4$, so capillarity is a modest but non-negligible correction. For $D = 10\,\mathrm{mm}$, $\mathrm{Bo} \approx 14$, and capillarity is safely negligible.

8.4 Flow regime and friction modelling

The Reynolds number

$$\mathrm{Re} = \frac{\rho v D}{\mu}$$

determines the flow regime. The Churchill correlation (11) spans all regimes, but the quasi-steady assumption (using instantaneous Re to evaluate f) is most accurate when velocity changes slowly relative to the viscous diffusion time D^2/ν.

For steady siphon flows in more detailed models, including direct calculation of the flow velocity and comparison with experiments, see Zheng et al. [9]. In IYPT-scale siphons, Reynolds numbers are often in the 10^3–10^4 range, so the flow is usually turbulent and well within the range where the Churchill correlation is most useful.

8.5 When the slug model breaks down

The single-slug approximation fails when:

- Strong three-dimensional separation or recirculation occurs (sharp bends, sudden expansions).
- Air is entrained at the inlet or outlet.
- The column separates (cavitation, trapped air pockets).
- Turbulent fluctuations are large compared to the mean flow.

In such regimes, volume-of-fluid CFD (e.g., OpenFOAM's `interFoam` [14] or ANSYS Fluent [15]) becomes necessary, and detailed CFD studies of siphon and inverted-siphon systems [10, 11] provide useful reference points. The slug model remains valuable as a diagnostic: systematic deviations from one-dimensional predictions highlight where additional physics is important.

8.6 Validity checklist

Before interpreting slug-model predictions quantitatively:

1. Verify the cavitation margin (32) is positive along the path to the crest.
2. Ensure Bo $\gg 1$ or apply a capillary correction.
3. Confirm typical Reynolds numbers lie in a regime where the friction model is appropriate.
4. Check that $|\dot{h}_c| \ll \sqrt{gD}$ (quasi-hydrostatic container).
5. Inspect experiments or CFD for signs of air entrainment or column separation.

9 Computational Implementation

9.1 Algorithm

The model is implemented as two sequential initial-value problems:

Filling phase. Rewrite (15) as a first-order system for (L, v):

$$\begin{aligned} \dot{L} &= v, \\ \dot{v} &= \frac{g}{L}\left[h_0 - \frac{A}{A_0}(L - L_1) - h(L)\right] - \frac{1}{2L}\left(f\frac{L}{D} + \sum K_i\right)|v|v. \end{aligned}$$

Integrate from $(L_1, 0)$ until the event $L = L_{\max}$ occurs at time t^*.

Draining phase. Integrate the system (28)–(29) from $(h_c(t^*), v(t^*))$ until $h_c \le h_{\text{out}}$ or $v \le 0$.

Standard adaptive Runge–Kutta methods (e.g., those in SciPy's `solve_ivp` [13] or the classical RK45) are adequate [12].

9.2 Python implementation

The complete, well-documented Python implementation used for all simulations in this work is provided externally in the script `simulation_code.py`, distributed alongside this manuscript. The code is developed and maintained outside the paper; the version bundled with this document should be used as the reference for reproducing the numerical results.

9.3 Ideal bound computation

The numerical routines that evaluate the ideal climb-height bound $h_{\max}$, including the helper function Φ and the root-finding procedure for the stall location, are implemented in `simulation_code.py`. These functions mirror the analytical expressions derived in Section 6 and are kept in the external code base to facilitate further development and reuse.

10 Model Summary

10.1 Key equations

Eq.	Expression	Meaning
(6)	$h_c(L) = h_0 - \dfrac{A}{A_0}(L - L_1)$	Container drawdown
(9)	$F_{\text{drive}} = \rho g A[h_c - h(L)]$	Driving force
(15)	$\ddot{L} = \dfrac{g}{L}[h_c - h(L)] - \dfrac{\text{losses}}{2L}\lvert\dot{L}\rvert\dot{L}$	Filling ODE
(18)	$\displaystyle\int_{L_1}^{L_{\text{stall}}} \frac{h_0 - h(\xi)}{\xi}\, d\xi = 0$	Ideal stall condition
(27)	$h_{\max} > h_3$	Priming criterion
(28)	$\dot{h}_c = -\dfrac{A}{A_0}v$	Draining mass balance
(29)	$\dot{v} = \dfrac{g}{L_{\max}}[h_c - h_{\text{out}}] - \dfrac{\text{losses}}{2L_{\max}}\lvert v\rvert v$	Draining momentum

10.2 Recommended workflow for IYPT investigations

1. **Specify geometry**: Define node coordinates $\{(x_i, h_i)\}$, compute segment lengths $\{L_i\}$, and construct $h(L)$.
2. **Check validity**: Apply the criteria of Section 8 to confirm the model regime.
3. **Compute ideal bound**: Evaluate $h_{\max}$ and check the priming criterion (27).
4. **Simulate full dynamics**: Integrate the filling and draining ODEs to obtain $L(t)$, $v(t)$, $h_c(t)$.
5. **Parameter studies**: Vary geometry, diameter, roughness, or A_0/A and record priming success, filling time, and draining rate.
6. **Compare with experiments**: Measure key observables and compare with model predictions; discrepancies identify missing physics.
7. **Use CFD selectively**: For borderline or anomalous cases, perform high-fidelity simulations to resolve three-dimensional effects.

11 Conclusions

This document has developed a one-dimensional slug-flow model for self-starting siphons, providing a theoretical and computational backbone for IYPT 2026 Problem #1.

The main contributions are:

- A geometric formulation via the piecewise-linear centreline function $h(L)$ and volume-conserving drawdown relation $h_c(L)$.
- A unified momentum derivation from first principles, yielding the driving force $F_{\text{drive}} = \rho g A[h_c - h(L)]$.
- An analytical upper bound $h_{\max}$ on climb height with a closed-form evaluation for piecewise-linear geometries.
- A geometric crest-crossing criterion $h_{\max} > h_3$ for rapid design screening.
- Complete filling and draining ODEs with Churchill friction and minor losses.

- Explicit validity criteria for cavitation, capillarity, flow regime, and inlet dynamics.
- A Python implementation suitable for parameter studies.

Future extensions could include an explicit cavitation switch, a meniscus-start model for small diameters, multi-slug tracking after column separation, and systematic comparison with experiments and CFD across a range of geometries.

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